

Flow: Connect the movement of Virtual Actor and Audience

Immersive Performance with Augmented Reality

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Figure 1: Screenshots of the AR scene, seen through the Flow AR application.

ABSTRACT

Composed of light, movement, and AR, <Flow> is an AR immersive performance in which the virtual actor and the audience create movement together. The virtual actor appears on stage through AR technology, can predict and flexibly respond to the audience's behavior based on AI. The audience interacts with the virtual actor through hand interaction designed based on 'Contact Improvisation'. The movements they form together are visualized on stage and tell a story of memory, while implicitly capturing the flow of relationships between two different worlds: virtual and reality, past and present, existence and non-existent.

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1 INTRODUCTION

<Flow> is an AR immersive performance in the form of physical theater where the virtual actor and the audience interact together. The virtual actor is a character that appears on stage through AR technology. Like [Billington and Shamash, 2018], the virtual actor interacts with the audience, adding their immersion in the story.

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The audience explores the movement by contacting the virtual actor. The flow of movement they create is visualized and tells the story about 'Memory'.

In <Flow>, the virtual actor elicits the audience's movements through 'Contact Improvisation', and the audience engages with the virtual actor, interacting and co-creating movements. 'Contact Improvisation' [Paxton, 1975] is a form of improvisational dance that involves continuous physical contact between bodies, allowing anyone to share their movements and form relationships regardless of identity. <Flow> utilizes these characteristics to design the interaction, unfolding a story of a person who lets go of memories and a person who holds onto them. As their movements are sketched across the space, it acquires multiple connotations.

We aim to advance the interaction with <Flow> system and to design an Audience Behavior Model in AR content by analyzing the collected audience data and the visualized movements. We expect <Flow> will serve as an important foundation for future AR content production and audience-centered interaction design.

2 EXPERIENCE DESCRIPTION

The stage of <Flow> is designed with several lights arranged (Figure 2). Each light symbolizes the memories shared between the key characters of <Flow>.

2.1 INTERACTION WITH VIRTUAL ACTOR

The virtual actor leads the audience's choreographic movements based on hand-based contact improvisation. The audience engages in Free-Hand interaction by directly touching the virtual actor with their hands, rather than using screen touch. The virtual actor and the audience move their joined hands, creating their unique flow of movements (Figure 2).

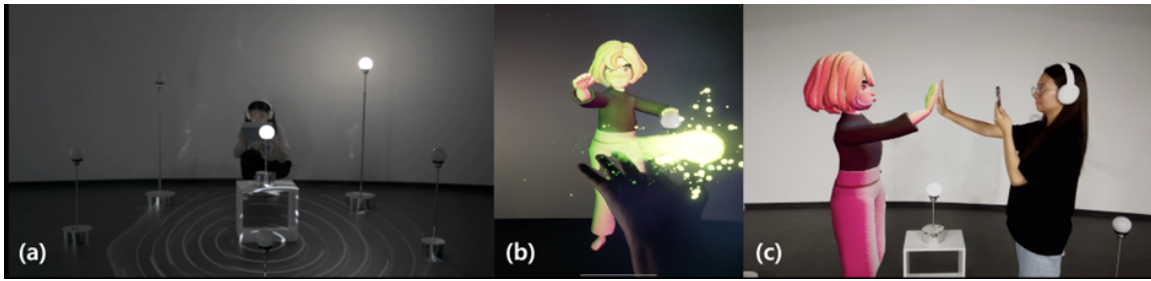


Figure 2: (a) Projection Mapping & Stage Design, (b, c) Examples of Free Hand Interaction. (First, Third Person View)

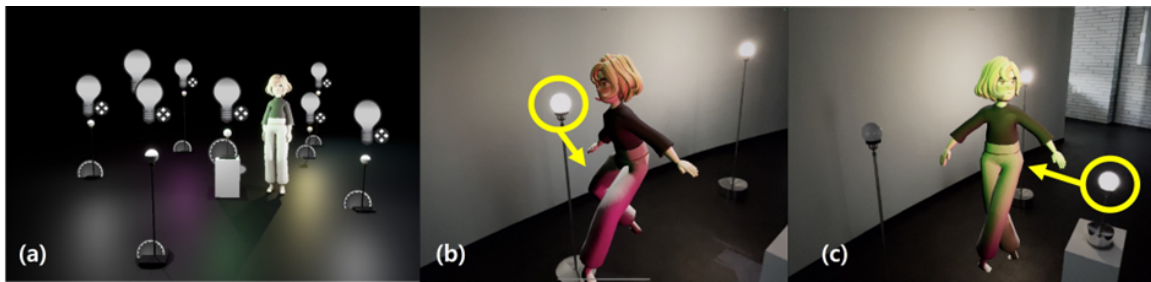


Figure 3: (a) Simulation view showing stage and AR set, (b, c) Physical Light and Virtual Light Interaction

The audience interacts with the virtual actor and freely moves between the lights on the stage. The movements created by the virtual actor and the audience gradually accumulate, generating multiple flows, which in turn affect the stage lighting. Their movements and the flickering of the lights show the flow of time, memories, and relationships.

2.2 MOVEMENT VISUALIZATION

<Flow> has a multi-layered experiential structure. The first-person view audience directly interacts with the virtual actor and primarily experiences hand-touch interaction with the virtual actor and the changing of the lights. The individual positional changes of the virtual actor and the audience, as well as the movements they create together, are visualized through projection mapping and AR elements. The third-person view audience can observe the visualized movement data of the key characters and discover the meaning of the relationships created between two different worlds: virtual and reality, past and present, existence and non-existence.

3 XR DEMONSTRATION

The main challenges and implementation details of <Flow> to be presented in the XR demonstration are as follows:

- **Flow AR App:** Developed using the AR application embedded in iPhone 14 Pro, iPad Pro, and Unity. We utilize LiDAR sensor-based AR technology and image recognition technology for user's Free-Hand interaction. The user's recognized hand is mapped in mesh form, and the audience interacts directly with the virtual actor by physically touching hands.
- **Interactive Virtual Actor:** In <Flow>, the virtual actor is an AI-based character that interacts with the physical set (lighting, projection mapping), virtual set (AR objects), and

the audience. The virtual actor predicts and flexibly interacts with the audience's actions, guiding their movements and responding differently based on the audience's reactions. Based on user experience surveys from previous AR works <Floating>, a behavior tree was constructed to guide the audience's interactions.

- **Collect and use real-time movement data:** The position information and movement data of the virtual actor and the audience are stored in real time and utilized for determining the virtual actor's actions and visualizing movement data.
- **Integrated Immersive System:** To create an immersive space where virtual and reality coexist, we have developed an immersive system that combines controllable lighting using Arduino, AR, projection mapping, and sound. By integrating virtual lighting with physical lighting, the virtual actor is also affected by the light when the physical lighting is turned on (Figure 3). Each element in the system operates by influencing each other based on audience data stored in real time, such as TeamLab's Forest of Resonating Lamps [TeamLab, 2020].

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